1. Title and Author Information

Title:
QSHIELD-FL: A Privacy-Preserving Quantum Federated Intrusion Detection System with Robust Aggregation

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2. Abstract

This project presents QSHIELD-FL, a privacy-preserving federated intrusion detection system based on a Hybrid Quantum Neural Network. The system combines quantum machine learning with federated learning to detect malicious network traffic without centralizing sensitive data. The NSL-KDD dataset is used for evaluation with proper preprocessing including one-hot encoding and feature scaling. The model is trained across multiple simulated clients and evaluated under clean, attacked, and defended scenarios. A SignFlip attack is introduced to simulate malicious clients in federated learning. Robust aggregation using Trimmed Mean and Differential Privacy noise injection are implemented to improve resilience and privacy. Experimental results show that clean federated learning achieves approximately 80% accuracy, while malicious updates reduce accuracy significantly. The proposed defense mechanisms successfully restore performance to near-clean levels. This work demonstrates the feasibility of combining quantum neural networks with secure federated learning for intrusion detection and sets the foundation for a hybrid classical-quantum extension in Phase-2.

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3. Keywords

Quantum Federated Learning; Hybrid Quantum Neural Network; Variational Quantum Circuit; Intrusion Detection System; Byzantine Robust Aggregation; Differential Privacy

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4. Introduction

Modern networks generate large volumes of distributed data. Intrusion Detection Systems (IDS) must detect malicious traffic patterns while maintaining privacy and robustness. Traditional centralized IDS models require aggregating data in one location, which may not be feasible in sensitive environments.

Federated Learning (FL) provides a solution by allowing distributed clients to train locally and share only model updates. However, FL systems are vulnerable to malicious participants who may send corrupted updates.

Quantum Machine Learning (QML) has emerged as a promising research direction, especially for handling high-dimensional feature representations and non-linear decision boundaries. However, integrating QML with federated and adversarial settings remains relatively unexplored.

This project proposes QSHIELD-FL, a Quantum Federated Learning framework for intrusion detection that:

- Uses a Hybrid Quantum Neural Network as the global model
- Simulates adversarial client behavior
- Applies robust aggregation
- Incorporates differential privacy

4.1 Research Gap

From the papers we read, we can see that most IDS research falls into separate tracks. Some papers focus mainly on high accuracy IDS models on datasets like NSL-KDD using deep learning and data balancing, but they are mostly centralized and do not consider how learning behaves when data is distributed across users.
Another set of works studies federated learning (FL) for intrusion detection and discusses design choices and challenges, but many of them do not deeply test the system under malicious client updates (Byzantine / poisoning) and do not integrate quantum models.
Separately, some papers propose robust aggregation or defenses (Trimmed Mean, Median-Krum, committee-based validation, backdoor defense), but they are not directly built as an end-to-end IDS pipeline that can be tested, compared, and demonstrated easily.
Also, privacy-focused FL works use DP, compression, or secure aggregation, but accuracy and robustness trade-offs are not always shown clearly in a security IDS setting.

So the gap is: very few works combine (1) intrusion detection + (2) federated learning + (3) adversarial client simulation + (4) robust aggregation + (5) differential privacy, and also explore this with a quantum model component.

QSHIELD-FL is built to address this gap by giving one integrated framework where we can run clean FL, attacked FL (SignFlip), defended FL (Trimmed Mean), and defended+DP, and then view results using a dashboard.

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5. Related Work

5.1 Classical IDS on NSL-KDD and Feature / Data Handling

Many IDS papers still use centralized training and focus on improving accuracy by better preprocessing and model design. A deep learning study on NSL-KDD and UNSW-NB15 shows that class imbalance is a big issue and resampling can change the detection results a lot, especially for rare attacks, so we should be careful while comparing accuracy.

Another work converts NSL-KDD features into image-like format (color mapping) so that CNN models can be used, and reports very high accuracy.
These works are useful because they show what accuracy can look like in centralized conditions, but they usually do not cover privacy risks of centralized data and do not test adversarial federated scenarios.

Connection to QSHIELD-FL: We also use NSL-KDD, but our focus is not only accuracy; we test robustness under malicious client updates and privacy mechanisms.

5.2 Quantum Machine Learning / Quantum Neural Networks for Learning Tasks

Quantum ML research often studies hybrid models where a classical network produces a small feature vector and a variational circuit does the final mapping. A quantum convolutional neural network (QCNN) based on variational circuits is one example that shows how VQC layers can be structured with encoding, trainable gates, and measurement.

There are also quantum-based security or intrusion-related works where a "quantum neural network" idea is used for detection and sometimes combined with optimization/encryption steps, but many of these are not tested under distributed learning and may be more "system-level" than FL-focused.

Connection to QSHIELD-FL: Our HQNN follows the standard hybrid pattern (classical feature layers + VQC + measurement + classical head), but we deploy it inside a federated pipeline with security testing.

5.3 Federated Learning for IDS

Federated learning is used to train IDS models while keeping data on client devices. A systematic review of FL-based IDS gives a taxonomy and discusses how distributed training helps privacy but also introduces new security problems like poisoning and model manipulation.

This direction motivates our design because IDS data is naturally distributed (labs, departments, different networks), so FL matches the real-world scenario better than centralized learning.

Connection to QSHIELD-FL: We do not only describe FL-IDS; we implement it with different experiment modes and show how accuracy changes under attack and defense.

5.4 Attacks and Defenses in Federated Learning (Byzantine / Backdoor / Robust Aggregation)

A major issue in FL is that a malicious client can send corrupted updates. Committee-based Byzantine-resilient FL proposes validation and reputation ideas to reduce the impact of attackers.

Backdoor defense papers show that even robust aggregation alone may fail in some poisoning settings, and defense should consider how attacks shift across parameters.
For robust aggregation, Trimmed Mean style rules are widely used because they drop extreme updates and keep the average stable.

Median-Krum is another Byzantine robust approach that uses distances/statistics to identify suspicious clients.

Connection to QSHIELD-FL: We simulate malicious behavior with a SignFlip-style attacker and show that naïve averaging degrades badly, then we show Trimmed Mean recovers accuracy close to clean training.

5.5 Privacy Mechanisms: Differential Privacy, Secure Aggregation, Compression, Leakage

Privacy in FL is not fully guaranteed just because raw data is not shared. There are gradient leakage attacks that reconstruct information from shared gradients, even when gradients are compressed.
Because of this, many works add differential privacy (DP) noise or secure aggregation. For example, some works build verifiable secure aggregation to prevent tampering and improve privacy trust.
Other papers combine compression + DP or chained FL structures to reduce overhead while keeping privacy.

Communication efficiency is also important in FL; works propose reputation-aware and distillation-based strategies to reduce communication costs for heterogeneous clients.

Connection to QSHIELD-FL: We include DP noise in our pipeline and show that it slightly reduces accuracy but still stays stable with defense. This helps demonstrate the privacy–utility tradeoff in a clear IDS setting.

5.6 Quantum Federated Learning (QFL) Context

Quantum federated learning is still a developing area and a recent literature review summarizes foundations, challenges (noise, resource limits, training stability), and open directions.
This supports our Phase-2 plan: instead of forcing “quantum for everything”, a hybrid approach (classical + quantum roles) may be more practical for accuracy and efficiency.

5.7 Comparison Table

Ref	Category	Dataset / Domain	Main Idea	Security / Privacy Focus	What it misses (gap)	Why it matters for QSHIELD-FL
[1]	Classical IDS	NSL-KDD, UNSW-NB15	Resampling + DL impact	No FL robustness	Centralized setting	motivates our NSL-KDD preprocessing + imbalance awareness

[2]	Quantum ML	Knowledge graphs	QML algorithm	Not security-focused	Not IDS/FL	shows how QML is structured and trained
[3]	FL Security	General FL	committee-based Byzantine resilience	Byzantine defense	Not IDS-specific	supports attacker/defense motivation
[4]	FL Security	General FL	backdoor defense beyond aggregation	Backdoor/poison	Not IDS pipeline	shows robust aggregation alone can fail
[5]	Robust Agg.	General FL	trimmed mean based FL framework	Byzantine defense	Not QML	matches our Trimmed Mean defense
[6]	Robust Agg.	General FL	Median-Krum	Byzantine defense	Not IDS/QML	alternative robust baseline for Phase-2 comparison
[7]	Robust+DP	General FL	robust aggregation compatible with DP	DP + robustness	Not IDS/QML	supports our defended+DP experiment
[8]	Privacy+Efficiency	General FL	quantization + chained FL	privacy + comm cost	Not IDS/QML	shows comm/privacy tradeoff
[9]	FL-IDS Survey	IDS	taxonomy of FL IDS	discusses poisoning	Survey (not system)	supports our research gap + design choices
[10]	Distributed IDS	satellite CAN traffic	distributed IDS	system security	Not QML/FL pipeline	shows distributed IDS is relevant beyond NSL-KDD
[11]	Classical IDS	NSL-KDD	hybrid SOM based IDS	No FL/DP	Centralized	baseline IDS style reference
[12]	QML (Hybrid)	healthcare	hybrid quantum-neural model	not FL security	not IDS	supports hybrid QNN

						pattern similar to ours
[13]	Secure Aggregation	FL networks	verifiable secure aggregation	privacy + integrity	not IDS/QML	supports future upgrades
[14]	QFL Survey	QFL	review of QFL	QFL challenges	no IDS	supports Phase-2 hybrid direction
[15]	Privacy attack	FL	leakage from compressed gradients	privacy threats	not IDS	motivates DP/noise and privacy mention
[16]	Classical IDS	NSL-KDD	CNN with color mapping	accuracy	no FL/robustness	shows centralized accuracy ceiling
[17]	Comm-efficient FL	general FL	compression/hierarchical FL	efficiency	not IDS/QML	future scalability discussion
[18]	Privacy+DP	FL	DP + compression approach	privacy	not IDS/QML	supports DP choices
[19]	Quantum circuit design	ML	QCNN / VQC design	not IDS	not FL	supports circuit design explanation
[20]	Quantum IDS system	IDS	quantum NN + optimization + ECC	security system	no FL/robustness	shows quantum IDS direction exists but lacks FL threat testing
[21]	Comm-efficient FL	FL	reputation/distillation efficiency	efficiency	not IDS/QML	future work for scaling QSHIELD-FL

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6. Problem Formulation

Let the dataset be:

$D = \{(x_i, y_i)\}$

where:

- $x_i \in \mathbb{R}^d$ represents the network feature vector
- $y_i \in \{0,1\}$ represents Normal or Attack

The objective is to learn a classifier f(x) that predicts y while satisfying:

1. Distributed training across multiple clients
2. Robustness against malicious updates
3. Privacy preservation
4. Stability over multiple federated rounds

In federated learning, each client k computes an update:

$\Delta W_k = W_k - W_global$

The server aggregates updates:

$W_new = W_global + Aggregation(\{\Delta W_k\})$

Aggregation may be:

- Standard Federated Averaging
- Trimmed Mean
- With or without DP noise

The final model must generalize well on unseen NSL-KDD test data.

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7. Proposed Method / Algorithm

7.1 System Architecture & Circuit Design

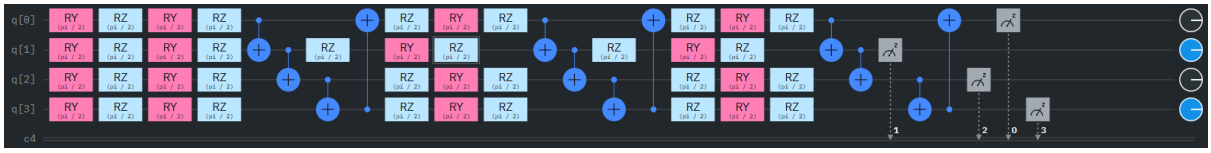


Figure 7.1.1 *QSHIELD-FL variational quantum circuit (4 qubits, 3 layers).* Features are encoded using $R_y(\cdot)$rotations, followed by three layers of $R_z - R_y - R_z$ trainable rotations and ring CNOT entanglement (0→1→2→3→0). Readout is based on $\langle Z \rangle$(shown as Z-basis measurement in Composer).

The overall workflow consists of:

1. NSL-KDD preprocessing
2. Client data partitioning
3. Local training using Hybrid Quantum Neural Network
4. Update aggregation at server
5. Attack simulation
6. Defense mechanism
7. Test evaluation

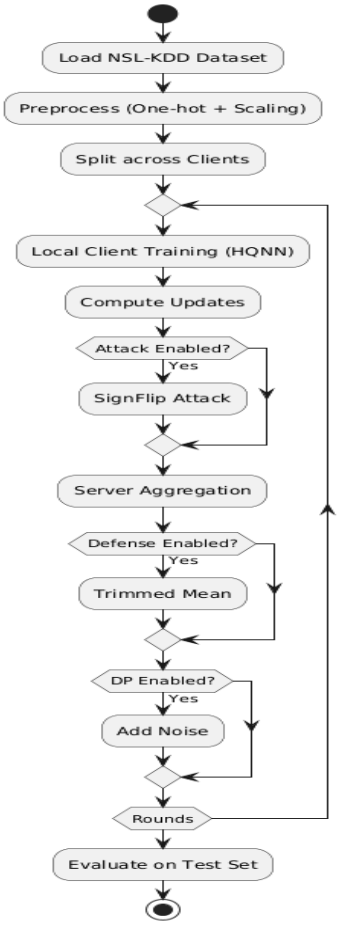


Figure 7.1.2 *QSHIELD-FL workflow (step-by-step).* Data is preprocessed, split across clients, trained locally using HQNN, then aggregated at the server. Optional modules include SignFlip attack simulation, Trimmed Mean defense, and DP noise addition, followed by evaluation on the test set.

7.2 Quantum Modelling

Q-Sphere

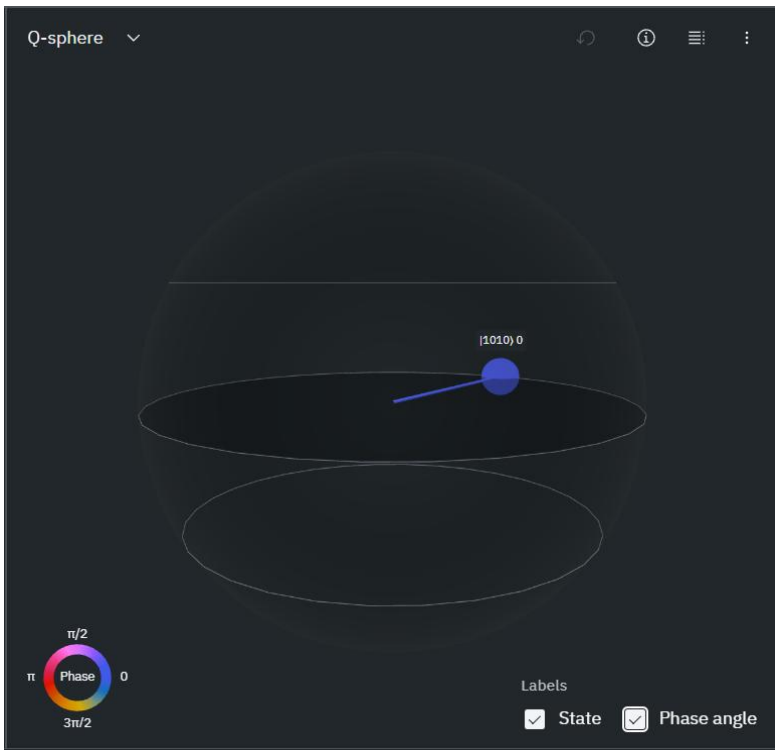


Figure 7.2.1 *Q-sphere visualization of the circuit output state.* It shows how the probability mass is distributed on the Bloch sphere representation, highlighting the dominant basis state(s) after running the VQC.

Probabilities

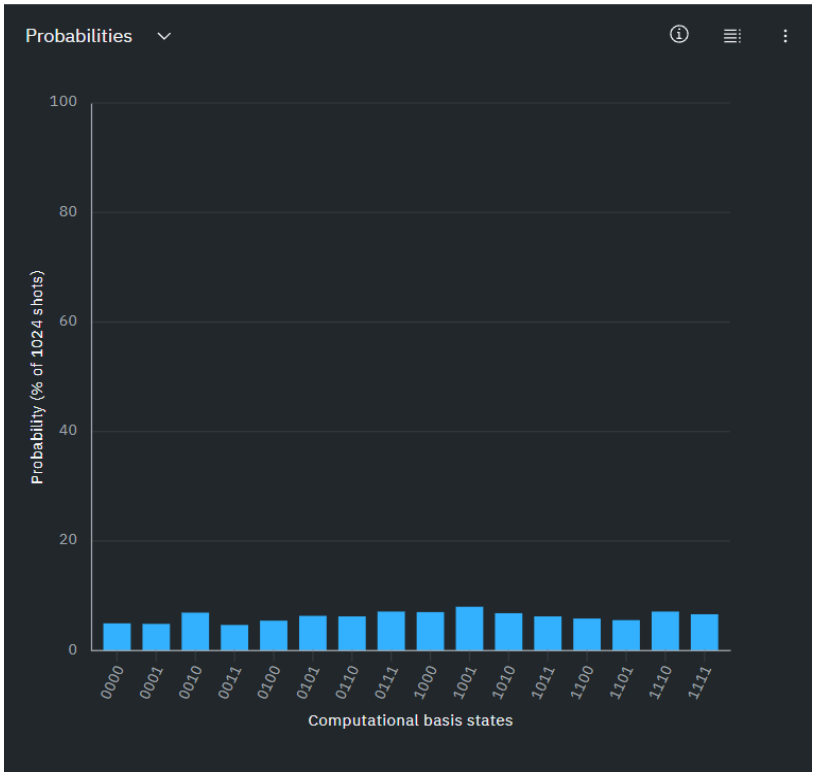


Figure 7.2.2 *Measurement probability distribution (1024 shots).* The bar chart shows the probabilities of different computational basis states obtained after executing the circuit.

Statevector

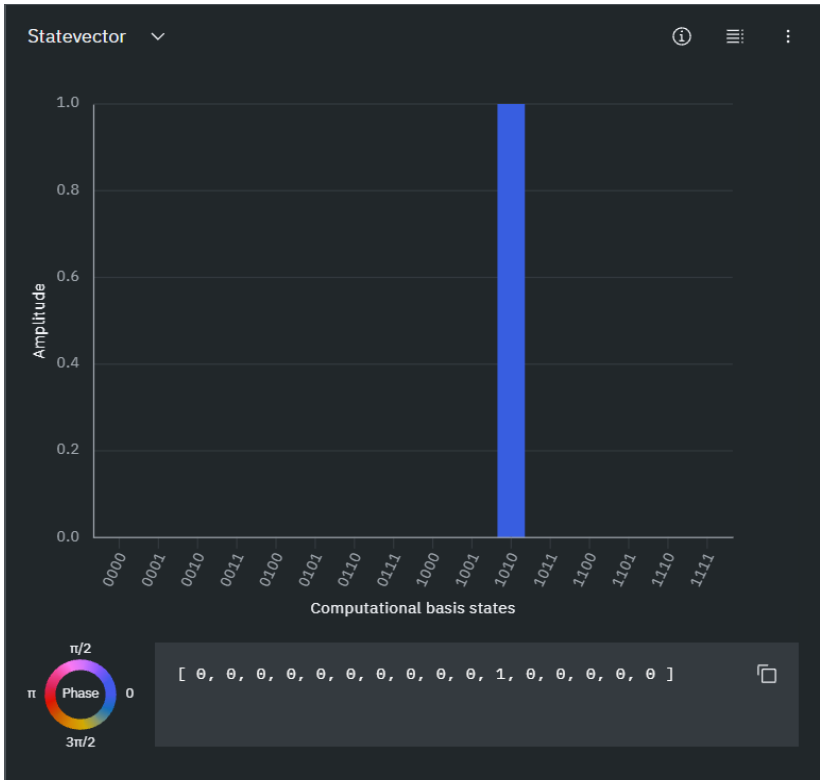


Figure 7.2.3 *Statevector amplitude plot (simulation).* The plot shows the amplitude of each computational basis state; one state has the highest amplitude, meaning the circuit output is concentrated there.

The core model is a Hybrid Quantum Neural Network implemented using PennyLane Lightning backend .

Architecture:

- Classical feature network (Linear → ReLU → Linear)
- AngleEmbedding with Y-rotations
- Multiple variational layers using Rot gates
- Ring CNOT entanglement
- PauliZ expectation value measurement
- Classical linear output head

Key implementation detail:

Features are not detached before passing to the quantum circuit, allowing full gradient propagation through the quantum node.

Resource awareness:

- 4 qubits
- 3 variational layers
- Statevector simulation (CPU optimized)

GPU was intentionally not used since Lightning simulator operates efficiently on CPU for small qubit counts.

7.3 Novelty and Contributions

The novelty of this work lies in combining multiple security-oriented mechanisms within a quantum federated setting. Compared to typical QML classification experiments, this system extends evaluation to adversarial and privacy contexts.

The specific contributions are:

1. Integration of a Hybrid Quantum Neural Network within a full federated learning pipeline for intrusion detection.
2. Implementation and evaluation of Byzantine SignFlip attack in a quantum federated environment.
3. Application of Trimmed Mean robust aggregation to restore performance under adversarial updates.
4. Integration of Differential Privacy noise injection into quantum federated training.
5. Development of a complete inference and analytics dashboard for reproducible evaluation.

Unlike many existing works that evaluate quantum models in centralized settings, this system studies quantum learning under realistic distributed and adversarial constraints.

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8. Experimental Setup

Dataset:

- NSL-KDD
- Binary classification (Normal vs Attack)

Environment:

- CPU execution only
- PennyLane Lightning backend
- PyTorch

Federated parameters:

- 3 clients (clean + attack experiments)
- 9 clients (defended experiments)
- 20 rounds
- 2 local epochs
- Learning rate: 3e-3
- Batch size: 256

Attack:

- SignFlip scale = 5.0

Defense:

- Trimmed Mean (trim_ratio = 0.25)

Privacy:

- Gaussian noise injection (sigma = 0.003)

Evaluation Metric:

- Test Accuracy
- Confusion matrix (via dashboard)

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9. Results and Comparative Analysis

(To be finalized in Phase-2, but current summary included.)

Observed final accuracies:

- Clean FL = 0.8012
- Attacked FL = 0.5035
- Defended FL = 0.7928

- Defended + DP = 0.7815

Analysis:

The SignFlip attack significantly degrades model performance, showing vulnerability of naive FedAvg. Trimmed Mean aggregation restores performance close to clean baseline. Adding Differential Privacy introduces slight degradation but maintains robustness.

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10. Discussion

The results demonstrate that:

- Quantum federated learning is feasible for intrusion detection.
- Federated models are vulnerable to malicious updates.
- Robust aggregation is necessary in adversarial environments.
- Privacy mechanisms introduce trade-offs but improve security guarantees.

Limitations:

- Binary classification only
- Simulation-based quantum backend
- Limited qubit count
- No classical baseline yet

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11. Conclusion

This work presents QSHIELD-FL, a privacy-preserving quantum federated intrusion detection system. The system successfully integrates quantum neural networks with federated learning, adversarial simulation, robust aggregation, and differential privacy.

Phase-2 will extend this system to include a hybrid classical-quantum architecture where a classical baseline model is trained and compared fairly against the quantum model, potentially enabling confidence-based gating mechanisms.

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12. References

[1] N. Dangol and A. Eaman, “Impact of resampling techniques in deep learning based intrusion detection: A comparative study on NSL-KDD and UNSW-NB15,” Procedia Computer Science, 2025, doi: 10.1016/j.procs.2025.10.182.

[2] Y. Ma and V. Tresp, “Quantum machine learning algorithm for knowledge graphs,” ACM Trans. Quantum Comput., vol. 2, 2021, doi: 10.1145/3467982.

[3] G. Xu et al., “CBRFL: A framework for committee-based Byzantine-resilient federated learning,” J. Netw. Comput. Appl., 2025, doi: 10.1016/j.jnca.2025.104165.

[4] G. Liu, W. Lin, T. Huang, F. Shi, X. Wang, and L. Shen, “Prodigal: Backdoor defense for federated learning beyond robust aggregation,” Knowledge-Based Systems, vol. 333, 2026, doi: 10.1016/j.knosys.2025.114903.

[5] T. Wang, Z. Zheng, and F. Lin, “Federated learning framework based on trimmed mean aggregation rules,” Expert Systems with Applications, vol. 270, 2025, doi: 10.1016/j.eswa.2024.126354.

[6] F. Colosimo and F. De Rango, “Median-Krum: A joint distance-statistical based Byzantine-robust algorithm in federated learning,” 2023, doi: 10.1145/3616390.3618283.

[7] K. Liu and K. Zhang, “A robust aggregation method compatible with differential privacy in federated learning,” 2025, doi: 10.1145/3778534.3778649.

[8] Y. Liu et al., “Quantization-based chained privacy-preserving federated learning,” Scientific Reports, 2025, doi: 10.1038/s41598-025-01420-5.

[9] J. L. Hernández-Ramos et al., “Intrusion detection based on federated learning: A systematic review,” ACM Computing Surveys, 2025, doi: 10.1145/3731596.

[10] O. Driouch, S. Bah, and Z. Guennoun, “CANSat-IDS: An adaptive distributed intrusion detection system for satellites, based on combined classification of CAN traffic,” Computers & Security, vol. 146, 2024, doi: 10.1016/j.cose.2024.104033.

[11] N. Iftikhar et al., “Intrusion detection in NSL-KDD dataset using hybrid self-organizing map model,” Computer Modeling in Engineering & Sciences, vol. 143, no. 1, 2025, doi: 10.32604/cmes.2025.062788.

[12] A. Verdone et al., “A hybrid quantum-neural network for heart disease classification,” Biomedical Signal Processing and Control, 2025, doi: 10.1016/j.bspc.2025.109185.

[13] W. Yao, T. Zhou, and Y. Han, “Verifiable secure aggregation scheme for privacy protection in federated learning networks,” Discover Computing, 2025, doi: 10.1007/s10791-025-09676-1.

[14] R. Ballester, J. Cerquides, and L. Artiles, “Quantum federated learning: A comprehensive literature review of foundations, challenges, and future directions,” Quantum Machine Intelligence, 2025, doi: 10.1007/s42484-025-00292-2.

[15] X. Ding et al., “Improved gradient leakage attack against compressed gradients in federated learning,” Neurocomputing, 2024, doi: 10.1016/j.neucom.2024.128349.

[16] M. A. Faiaz, “Intrusion detection using convolutional neural network: A color mapping approach on NSL-KDD dataset,” 2024, doi: 10.1145/3704522.3704541.

[17] Y. Liu et al., “HierFLMC: Efficient hierarchical federated learning based on soft clustering model compression,” Expert Systems with Applications, 2025, doi: 10.1016/j.eswa.2025.130022.

[18] A. El Ouadrhiri et al., “Privacy-preserving federated learning approach based on Hensel’s compression and differential privacy,” Discover Internet of Things, 2025, doi: 10.1007/s43926-025-00236-z.

[19] L. H. Gong et al., “Quantum convolutional neural network based on variational quantum circuits,” Optics Communications, 2024, doi: 10.1016/j.optcom.2023.129993.

[20] H. Kadry, A. Farouk, E. A. Zanaty, and O. Reyad, “Intrusion detection model using optimized quantum neural network and elliptical curve cryptography for data security,” Alexandria Engineering Journal, vol. 71, pp. 491–500, 2023, doi: 10.1016/j.aej.2023.03.072.

[21] Y. Li et al., “Communication-efficient federated learning for heterogeneous clients,” ACM Trans. Internet Technol., 2025, doi: 10.1145/3716870.